index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Experiment 1 - Matomo Analytics</title>
  <link rel="stylesheet" href="style.css" />

  <!-- Matomo -->
  <script>
    var _paq = (window._paq = window._paq || []);
    /* tracker methods like "setCustomDimension" should be called before "trackPageView" */
    _paq.push(["trackPageView"]);
    _paq.push(["enableLinkTracking"]);
    (function () {
      var u = "//localhost/matomo/";
      _paq.push(["setTrackerUrl", u + "matomo.php"]);
      _paq.push(["setSiteId", "1"]);
      var d = document,
        g = d.createElement("script"),
        s = d.getElementsByTagName("script")[0];
      g.async = true;
      g.src = u + "matomo.js";
      s.parentNode.insertBefore(g, s);
    })();
  </script>
  <!-- End Matomo Code -->
</head>
<body>
  <main class="container">
    <h1>Experiment 1: Web Analytics Using Matomo</h1>
    <p>
      This page demonstrates how to integrate Matomo, an open-source analytics tool.
      Matomo tracking is configured for <strong>localhost/matomo</strong> with Site ID <strong>1</strong>.
      Update these values in the tracking snippet if your Matomo setup is different.
    </p>

    <section>
      <h2>What This Tracks</h2>
      <ul>
        <li>Page views</li>
        <li>Visit duration</li>
        <li>Custom button click event below</li>
      </ul>
    </section>

    <button id="trackButton" type="button">Track Sample Event</button>
    <p id="status" aria-live="polite"></p>

    <p>Devang Deokule - A58</p>
  </main>

  <script src="script.js"></script>
</body>
</html>
```

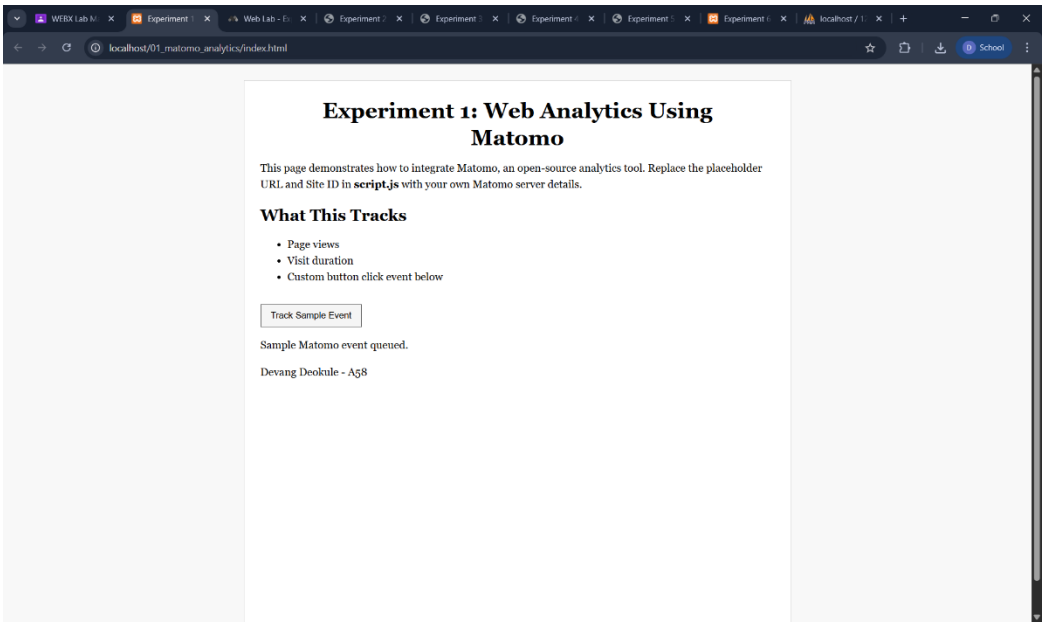
script.js:

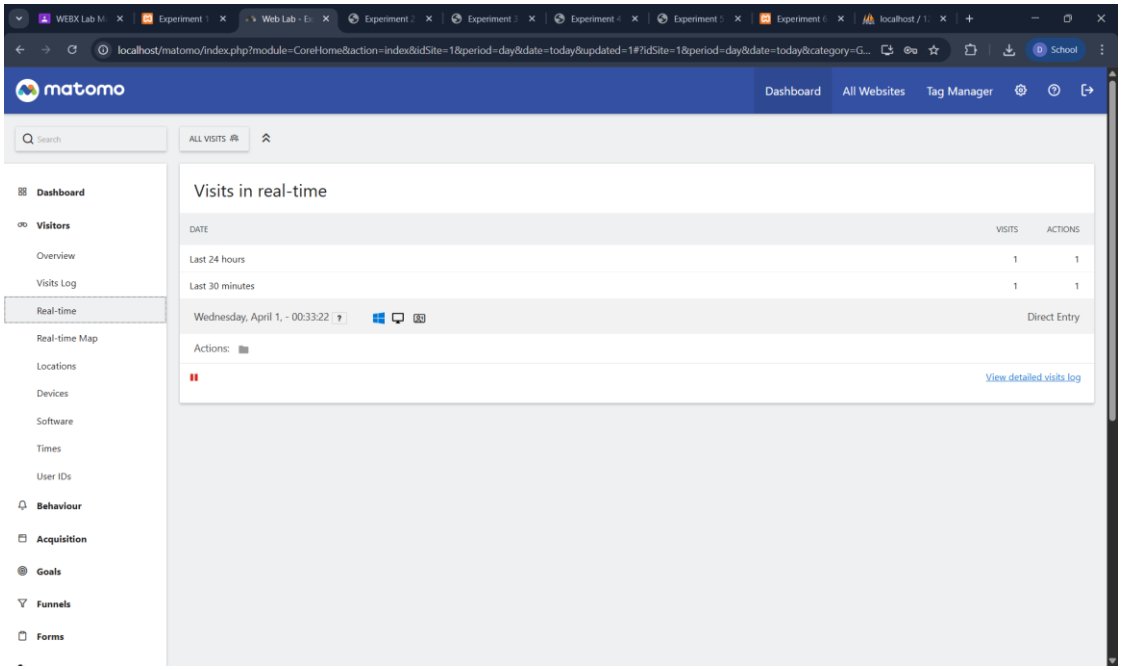
```
var trackButton = document.getElementById("trackButton");

if (trackButton) {
  trackButton.addEventListener("click", function () {
    var paq = (window._paq = window._paq || []);
    paq.push(["trackEvent", "Experiment", "ButtonClick", "Sample Event"]);
    var status = document.getElementById("status");
    if (status) {
      status.textContent = "Sample Matomo event queued.";
    }
  });
}
```

style.css:

```
body {
  margin: 0;
  font-family: Georgia, "Times New Roman", serif;
  line-height: 1.5;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  padding: 1.5rem 1rem;
  background: #f8f8f8;
}
.container {
  width: min(100%, 760px);
  padding: 1.5rem;
  border: 1px solid #d8d8d8;
  background: #fff;
}
h1,
h2 {
  margin: 0 0 0.65rem;
}
h1 {
  max-width: 28ch;
  margin-left: auto;
  margin-right: auto;
  text-align: center;
  font-weight: 700;
  line-height: 1.25;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}
button {
  border: 1px solid #444;
  background: #f5f5f5;
  padding: 0.5rem 0.9rem;
  cursor: pointer;
}
#status {
  margin-top: 0.9rem;
  min-height: 1.5rem;
}
section,
ul,
p,
button {
  margin-top: 0.8rem;
```

Output:





index.html:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
  <title>Experiment 2 - TypeScript Calculator</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <main class="container">
    <h1>Experiment 2: Calculator in TypeScript</h1>

    <div class="row">
      <label for="num1">First Number</label>
      <input id="num1" type="number" value="10" />
    </div>

    <div class="row">
      <label for="num2">Second Number</label>
      <input id="num2" type="number" value="5" />
    </div>

    <div class="row">
      <label for="operator">Operator</label>
      <select id="operator">
        <option value="+">+</option>
        <option value="-">-</option>
        <option value="*">*</option>
        <option value="/">/</option>
      </select>
    </div>

    <button id="calculateBtn"
type="button">Calculate</button>
    <p id="result"></p>

    <p>Devang Deokule - A58</p>
  </main>

  <script src="calculator.js"></script>
</body>
</html>
```

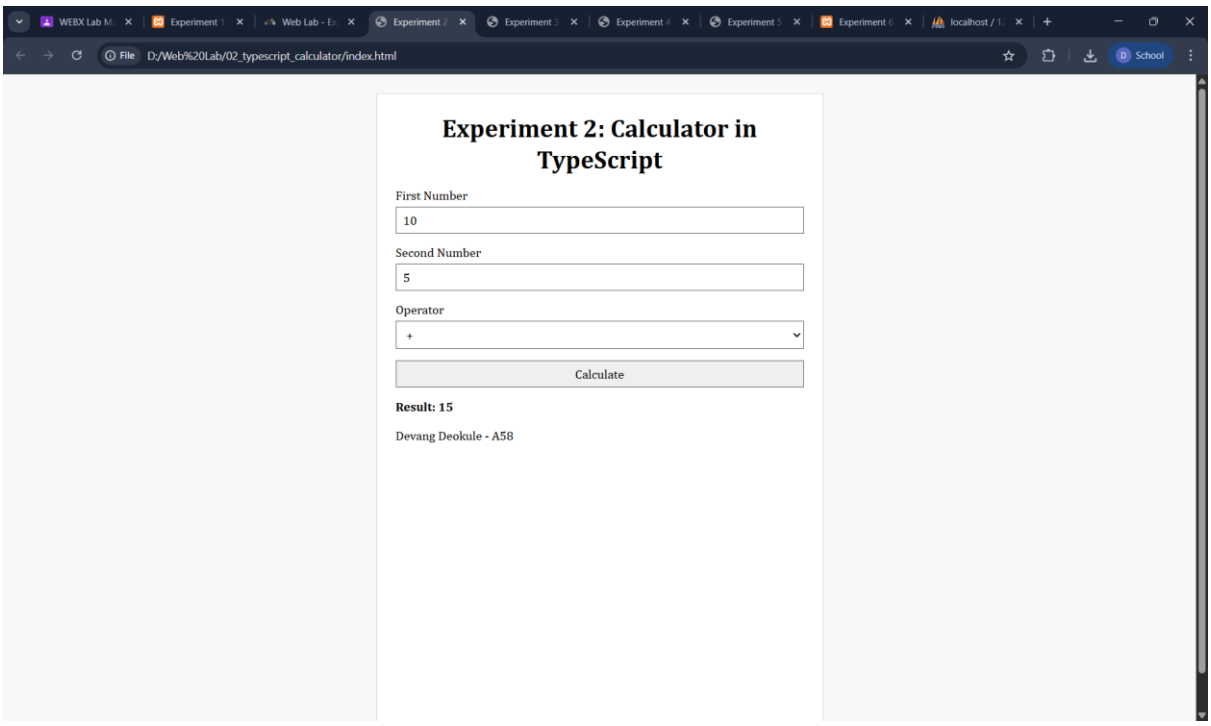
calculator.js:

```js
var Calculator = /** @class */ (function () {
  function Calculator() {}
  Calculator.prototype.compute = function (a, b, op) {
    switch (op) {
      case "+":
        return a + b;
      case "-":
        return a - b;
      case "*":
        return a * b;
      case "/":
        if (b === 0) {
          throw new Error("Division by zero is not allowed.");
        }
        return a / b;
    }
  };
  return Calculator;
})();

var calculator = new Calculator();

var calculateBtn =
document.getElementById("calculateBtn");
var resultNode = document.getElementById("result");

calculateBtn.addEventListener("click", function () {
  var num1 =
Number(document.getElementById("num1").value);
  var num2 =
Number(document.getElementById("num2").value);
  var operator =
document.getElementById("operator").value;

  try {
    var result = calculator.compute(num1, num2, operator);
    resultNode.textContent = "Result: " + result;
  } catch (error) {
    resultNode.textContent = error && error.message ?
error.message : "Unexpected error.";
  }
});
```

styles.css:

```
body {
  margin: 0;
  font-family: Cambria, Cochin, Georgia, serif;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  padding: 1.5rem 1rem;
  background: #f8f8f8;
}

.container {
  width: min(100%, 520px);
  padding: 1.5rem;
  border: 1px solid #d8d8d8;
  background: #fff;
}

.row {
  margin: 0.9rem 0;
}

h1 {
  max-width: 26ch;
  margin: 0 auto 1rem;
  text-align: center;
  font-weight: 700;
  line-height: 1.25;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}

label {
  display: block;
  margin-bottom: 0.35rem;
}

input,
select,
button {
  font: inherit;
  width: 100%;
  padding: 0.45rem 0.55rem;
  border: 1px solid #555;
  box-sizing: border-box;
}

button {
  background: #efefef;
  cursor: pointer;
}

#result {
  margin-top: 0.95rem;
  font-weight: 600;
  min-height: 1.3rem;
}
```

Output:



index.html:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Experiment 3 - TypeScript Multilevel Inheritance</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <main class="container">
    <h1>Experiment 3: Multilevel Inheritance in TypeScript</h1>
    <p>This example shows inheritance chain: Person - Student - GraduateStudent.</p>
    <button id="showBtn" type="button">Show Output</button>
    <pre id="output"></pre>
    <p>Devang Deokule - A58</p>
  </main>

  <script src="inheritance.js"></script>
</body>
</html>
```

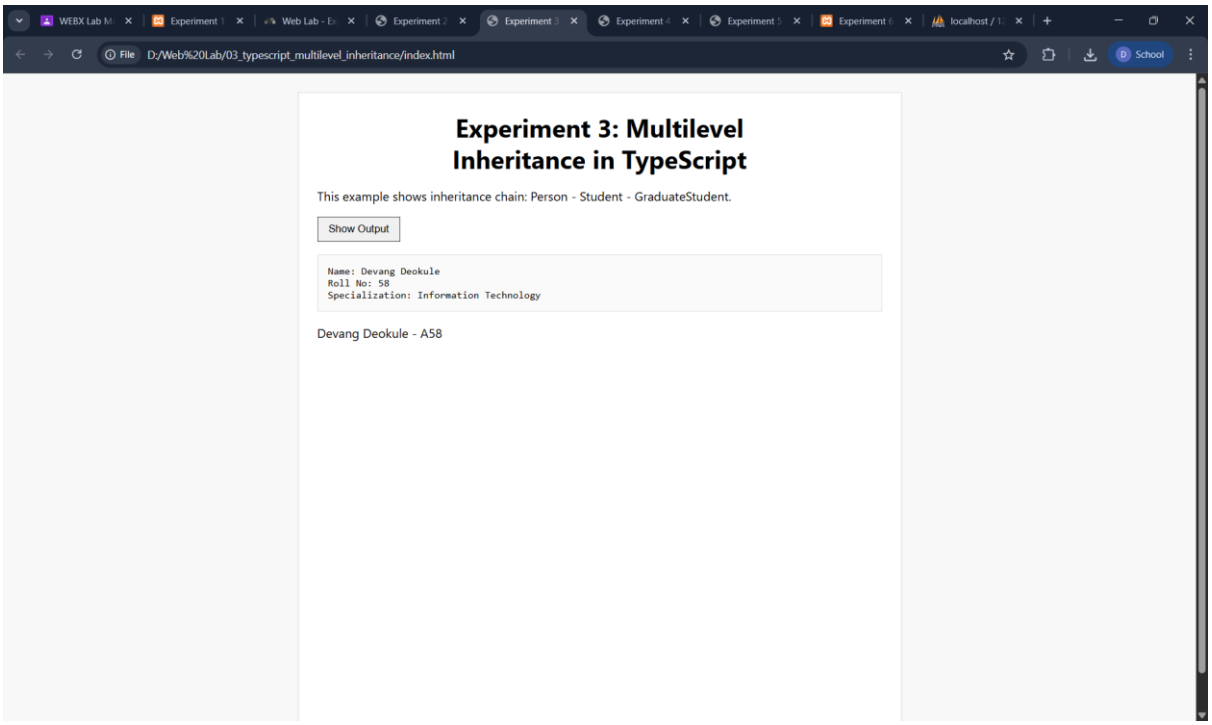
inheritance.ts:

```ts
export {};

class Person {
  constructor(protected name: string) {}

  public describe(): string {
    return `Name: ${this.name}`;
  }
}

class Student extends Person {
  constructor(name: string, protected rollNo: number) {
    super(name);
  }

  public describe(): string {
    return `${super.describe()}\nRoll No: ${this.rollNo}`;
  }
}

class GraduateStudent extends Student {
  constructor(name: string, rollNo: number, private specialization: string) {
    super(name, rollNo);
  }

  public describe(): string {
    return `${super.describe()}\nSpecialization: ${this.specialization}`;
  }
}

document.getElementById("showBtn")?.addEventListener("click", () => {
  const graduate = new GraduateStudent("Devang Deokule", 58, "Information Technology");
  const output = document.getElementById("output") as HTMLPreElement;
  output.textContent = graduate.describe();
});
```

styles.css:

```
body {
  margin: 0;
  font-family: "Segoe UI", Tahoma, Geneva,
Verdana, sans-serif;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  padding: 1.5rem 1rem;
  background: #f8f8f8;
}

.container {
  width: min(100%, 720px);
  padding: 1.5rem;
  border: 1px solid #d8d8d8;
  background: #fff;
}

h1 {
  max-width: 28ch;
  margin: 0 auto 1rem;
  text-align: center;
  font-weight: 700;
  line-height: 1.25;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}

button {
  border: 1px solid #444;
  background: #f0f0f0;
  padding: 0.45rem 0.85rem;
  cursor: pointer;
}

pre {
  margin-top: 1rem;
  border: 1px solid #ddd;
  padding: 0.8rem;
  background: #fafafa;
  white-space: pre-wrap;
}
```

Output:



index.html:

```html
<!DOCTYPE html>
<html lang="en" ng-app="basicSpaApp">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Experiment 4 - AngularJS SPA Basic</title>
  <link rel="stylesheet" href="styles.css" />
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular.min.js"></script>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/angular-route.min.js"></script>
</head>
<body>
  <main class="container">
    <h1>Experiment 4: AngularJS Single Page Application</h1>
    <nav>
      <a href="#!/home">Home</a>
      <a href="#!/about">About</a>
      <a href="#!/contact">Contact</a>
    </nav>
    <section ng-view></section>
    <p>Devang Deokule - A58</p>
  </main>

  <script src="app.js"></script>
</body>
</html>
```

app.js:

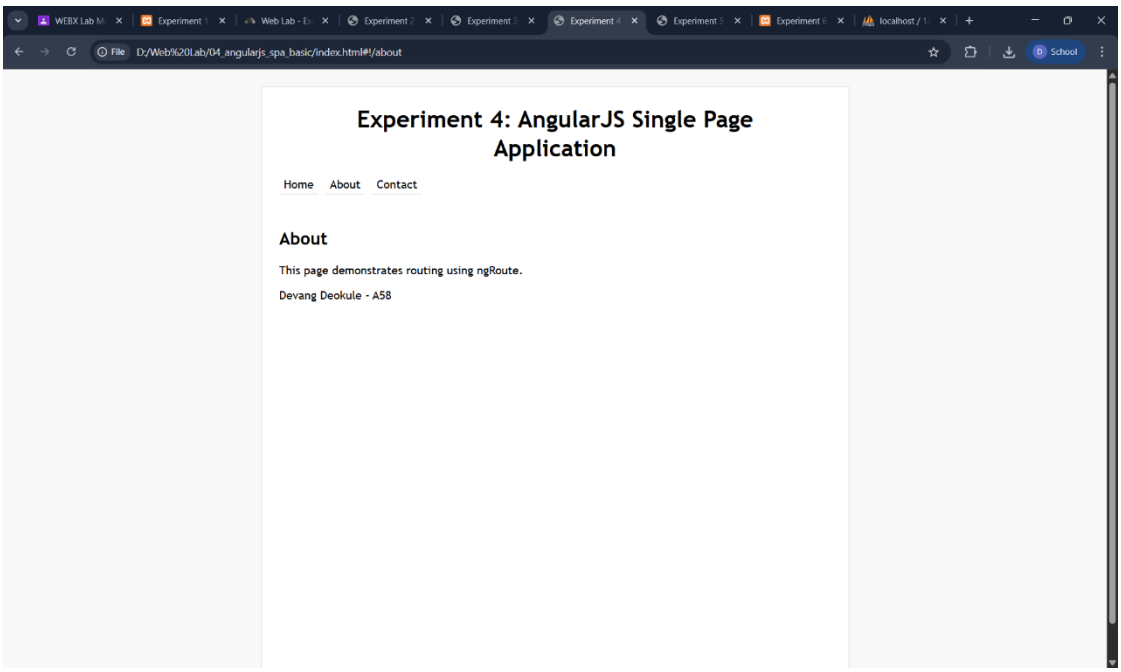
```javascript
var app = angular.module("basicSpaApp", ["ngRoute"]);
app.config(function ($routeProvider) {
  $routeProvider
    .when("/home", {
      template: "<h2>Home</h2><p>Welcome to the AngularJS SPA home page.</p>"
    })
    .when("/about", {
      template: "<h2>About</h2><p>This page demonstrates routing using ngRoute.</p>"
    })
    .when("/contact", {
      template: "<h2>Contact</h2><p>Email: deokuledevang@gmail.com</p>"
    })
    .otherwise({
      redirectTo: "/home"
    });
});
```

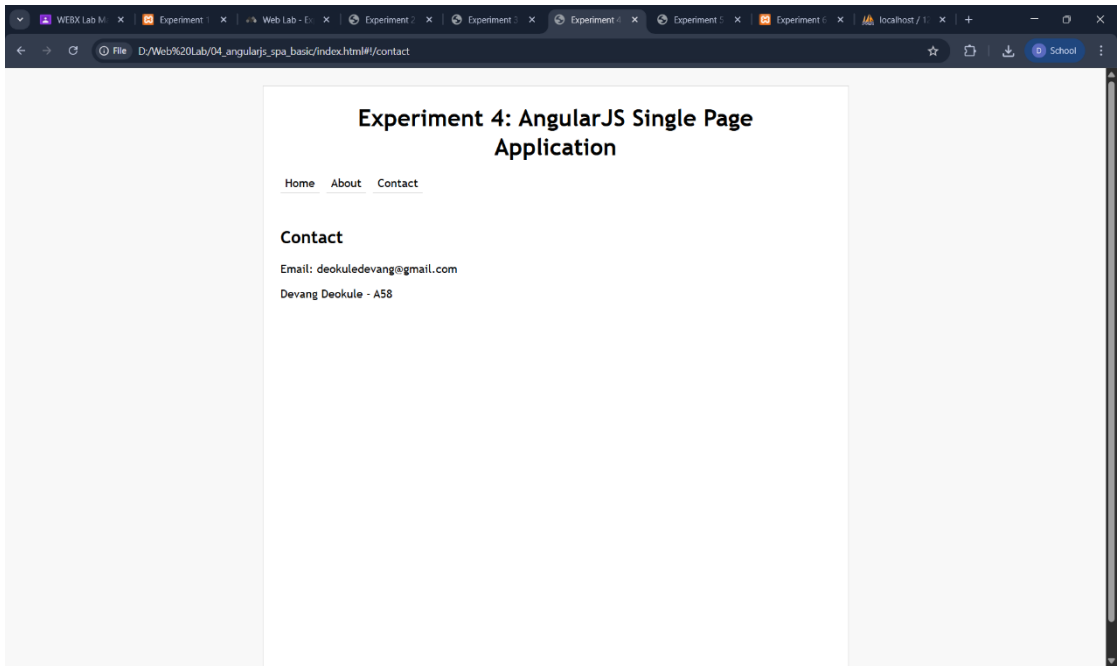
styles.css:

```css
body {
  margin: 0;
  font-family: "Trebuchet MS", sans-serif;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  padding: 1.5rem 1rem;
  background: #f8f8f8;
}
.container {
  width: min(100%, 760px);
  padding: 1.5rem;
  border: 1px solid #d8d8d8;
  background: #fff;
```

```
}
h1 {
  max-width: 30ch;
  margin: 0 auto 1rem;
  text-align: center;
  font-weight: 700;
  line-height: 1.25;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}
nav {
  margin: 1rem 0;
  display: flex;
  flex-wrap: wrap;
  gap: 0.6rem;
}
nav a {
  margin-right: 0;
  padding: 0.25rem 0.4rem;
  text-decoration: none;
  color: inherit;
  border-bottom: 1px solid #ccc;
}
[ng-view] {
  padding-top: 0.7rem;
}
```

Output:





index.html:

```
<!DOCTYPE html>
<html lang="en" ng-app="advancedSpaApp">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
  <title>Experiment 5 - AngularJS Services Events
Validation Helpers</title>
  <link rel="stylesheet" href="styles.css" />
  <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.3/an
gular.min.js"></script>
</head>
<body ng-controller="MainController as vm">
  <main class="container">
    <h1>Experiment 5: AngularJS SPA with Services, Events,
Validation, Helpers</h1>

    <form name="taskForm" novalidate ng-
submit="vm.addTask(taskForm)">
      <label for="taskInput">Task</label>
      <input
        id="taskInput"
        name="taskInput"
        type="text"
        ng-model="vm.newTask"
        ng-minlength="3"
        required
      />
      <small ng-show="taskForm.taskInput.$touched &&
taskForm.taskInput.$invalid">
        Task is required and must be at least 3 characters.
      </small>
      <button type="submit">Add Task</button>
    </form>

    <p><strong>Total Tasks:</strong> {{ vm.taskCount()
}}</p>

    <ul>
      <li ng-repeat="item in vm.tasks track by $index">
        {{ item.text }}
        <button type="button" ng-
click="vm.removeTask($index)">Remove</button>
      </li>
    </ul>

    <p aria-live="polite">{{ vm.message }}</p>
    <p>Devang Deokule - A58</p>
  </main>

  <script src="app.js"></script>
</body>
</html>
```

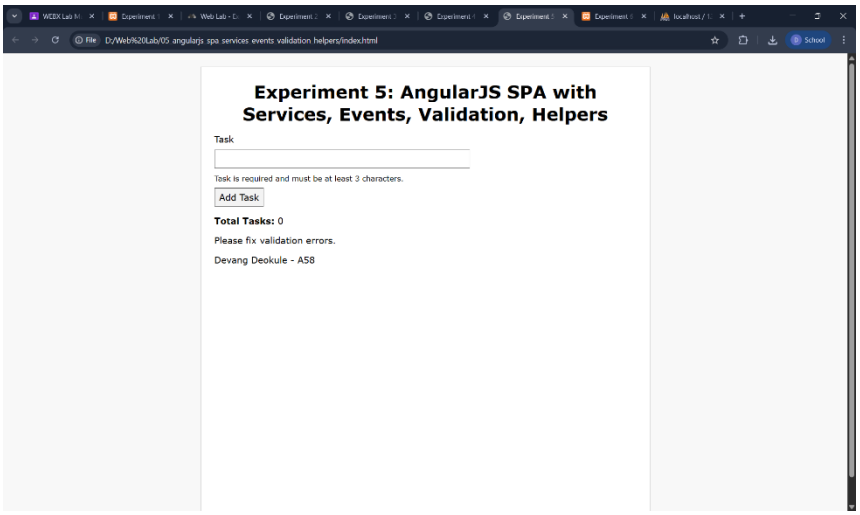
app.js:

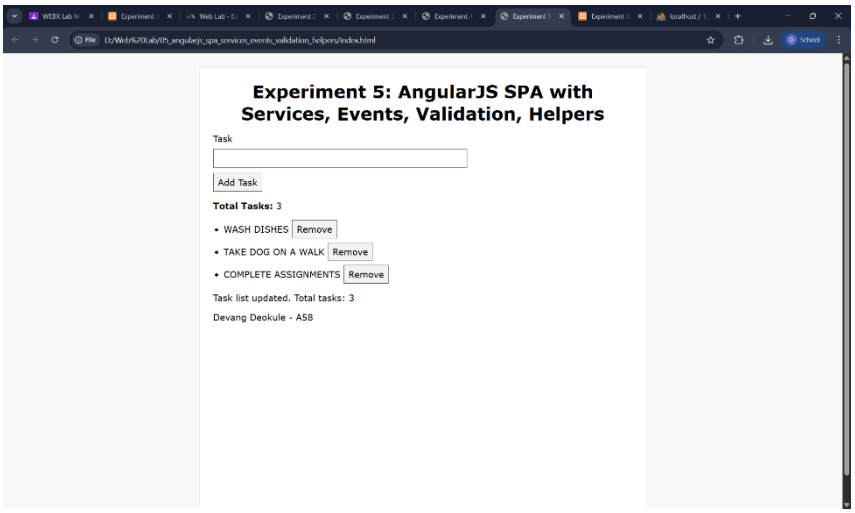
```
var app = angular.module("advancedSpaApp", []);
app.service("TaskService", function ($rootScope) {
  var tasks = [];
  this.list = function () {
    return tasks;
  };
  this.add = function (text) {
    tasks.push({ text: text.trim() });
    $rootScope.$broadcast("task:updated", tasks.length);
  };
  this.remove = function (index) {
    tasks.splice(index, 1);
    $rootScope.$broadcast("task:updated", tasks.length);
  };
});
app.controller("MainController", function ($scope,
TaskService, $filter) {
  var vm = this;
  vm.tasks = TaskService.list();
  vm.newTask = "";
  vm.message = "Add a task to begin.";
  vm.addTask = function (form) {
    if (form.$invalid) {
      vm.message = "Please fix validation errors.";
      return;
    }
    var cleanedTask =
$filter("uppercase")(vm.newTask.trim());
    TaskService.add(cleanedTask);
    vm.newTask = "";
    form.$setPristine();
    form.$setUntouched();
  };
  vm.removeTask = function (index) {
    TaskService.remove(index);
  };
  vm.taskCount = function () {
    return vm.tasks.length;
  };
  $scope.$on("task:updated", function (_event, count) {
    vm.message = "Task list updated. Total tasks: " + count;
  });
})
```

styles.css:

```
body {
  margin: 0;
  font-family: Verdana, Geneva, Tahoma, sans-serif;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  padding: 1.5rem 1rem;
  background: #f8f8f8;
}
.container {
  width: min(100%, 760px);
  padding: 1.5rem;
  border: 1px solid #d8d8d8;
  background: #fff;
}
h1 {
  max-width: 34ch;
  margin: 0 auto 1rem;
  text-align: center;
  font-weight: 700;
  line-height: 1.25;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}
form {
  margin-bottom: 1rem;
  display: grid;
  gap: 0.6rem;
  max-width: 460px;
}
input,
button {
  font: inherit;
  padding: 0.4rem 0.5rem;
  border: 1px solid #555;
}
input {
  width: 100%;
  box-sizing: border-box;
}
button {
  margin-left: 0;
  background: #f3f3f3;
  width: fit-content;
}
small {
  display: block;
  margin: 0;
}
ul {
  margin-top: 0.9rem;
  padding-left: 1.2rem;
}
li {
  margin: 0.45rem 0;
}
```

Output:





index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Experiment 6 - AJAX Search</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <main class="container">
    <h1>Experiment 6: AJAX Based Search</h1>
    <label for="searchInput">Search Users</label>
    <input id="searchInput" type="search" placeholder="Type a name..." />
    <ul id="results"></ul>
    <p>Devang Deokule - A58</p>
  </main>
  <script src="app.js"></script>
</body>
</html>
```

app.js:

```
var input = document.getElementById("searchInput");
var results = document.getElementById("results");
function parseXmlResponse(request) {
  var xml = request.responseXML;
  if (xml &&
!xml.getElementsByTagName("parsererror").length) {
    return { xml: xml };
  }
  var raw = request.responseText ?
request.responseText.trim() : "";
  if (!raw) {
    return { error: "Empty response from PHP backend." };
  }
  if (raw.indexOf("<?php") !== -1) {
    return {
      error:
        "PHP code was returned instead of XML. Run this
experiment on a PHP server.",
    };
  }
  var parsed = new DOMParser().parseFromString(raw,
"application/xml");
  if (parsed.getElementsByTagName("parsererror").length) {
    return { error: "Backend response is not valid XML." };
  }
  return { xml: parsed };
}
function searchUsers(query) {
  var request = new XMLHttpRequest();
  request.open("GET", "search.php?q=" +
encodeURIComponent(query), true);
  request.onreadystatechange = function () {
    if (request.readyState !== 4) {
      return;
    }
    if (request.status !== 200) {
      results.innerHTML = "<li>Unable to fetch data from
PHP backend.</li>";
      return;
    }
    var parsed = parseXmlResponse(request);
    if (!parsed.xml) {
      results.innerHTML = "<li>" + parsed.error + "</li>";
      return;
    }

    var xml = parsed.xml;
    var errorNode = xml.getElementsByTagName("error")[0];
    if (errorNode) {
      results.innerHTML = "<li>" + errorNode.textContent +
"</li>";
      return;
    }
    var users = xml.getElementsByTagName("user");
    render(users);
  };
  request.send();
}
input.addEventListener("input", function () {
  searchUsers(input.value.trim());
});
function render(userNodes) {
  if (!userNodes.length) {
    results.innerHTML = "<li>No matching users.</li>";
```

```
    return;
  }
  var html = "";
  for (var i = 0; i < userNodes.length; i++) {
    var name =
userNodes[i].getElementsByTagName("name")[0].textConte
nt;
    var email =
userNodes[i].getElementsByTagName("email")[0].textCont
ent;
    html += "<li>" + name + " (" + email + ")</li>";
  }
  results.innerHTML = html;
}
searchUsers("");
```

search.php:

```
<?php
declare(strict_types=1);
header('Content-Type: application/xml; charset=UTF-8');
$query = isset($_GET['q']) ? trim((string) $_GET['q']) : '';
$xmlPath = __DIR__ . '/users.xml';
function xmlEscape(string $value): string
{
    return htmlspecialchars($value, ENT_XML1 |
ENT_QUOTES, 'UTF-8');
}
if (!file_exists($xmlPath)) {
    http_response_code(500);
    echo "<?xml version=\"1.0\" encoding=\"UTF-8\"?>\n";
    echo "<results><error>Backend XML file not
found.</error></results>";
    exit;
}
if (!function_exists('simplexml_load_file')) {
    http_response_code(500);
    echo "<?xml version=\"1.0\" encoding=\"UTF-8\"?>\n";
    echo "<results><error>SimpleXML extension is not
enabled in PHP.</error></results>";
    exit;
}
libxml_use_internal_errors(true);
$users = simplexml_load_file($xmlPath);
if ($users === false) {
    libxml_clear_errors();
    http_response_code(500);
    echo "<?xml version=\"1.0\" encoding=\"UTF-8\"?>\n";
    echo "<results><error>Unable to read backend XML
file.</error></results>";
    exit;
}
libxml_clear_errors();
$response = "<?xml version=\"1.0\" encoding=\"UTF-
8\"?>\n<results>";
foreach ($users->user as $user) {
    $name = trim((string) $user->name);
    $email = trim((string) $user->email);
    if ($query !== '' && stripos($name, $query) === false) {
        continue;
    }
    $response .= '<user>';
    $response .= '<name>' . xmlEscape($name) . '</name>';
    $response .= '<email>' . xmlEscape($email) . '</email>';
    $response .= '</user>';
}
$response .= '</results>';
echo $response;
```

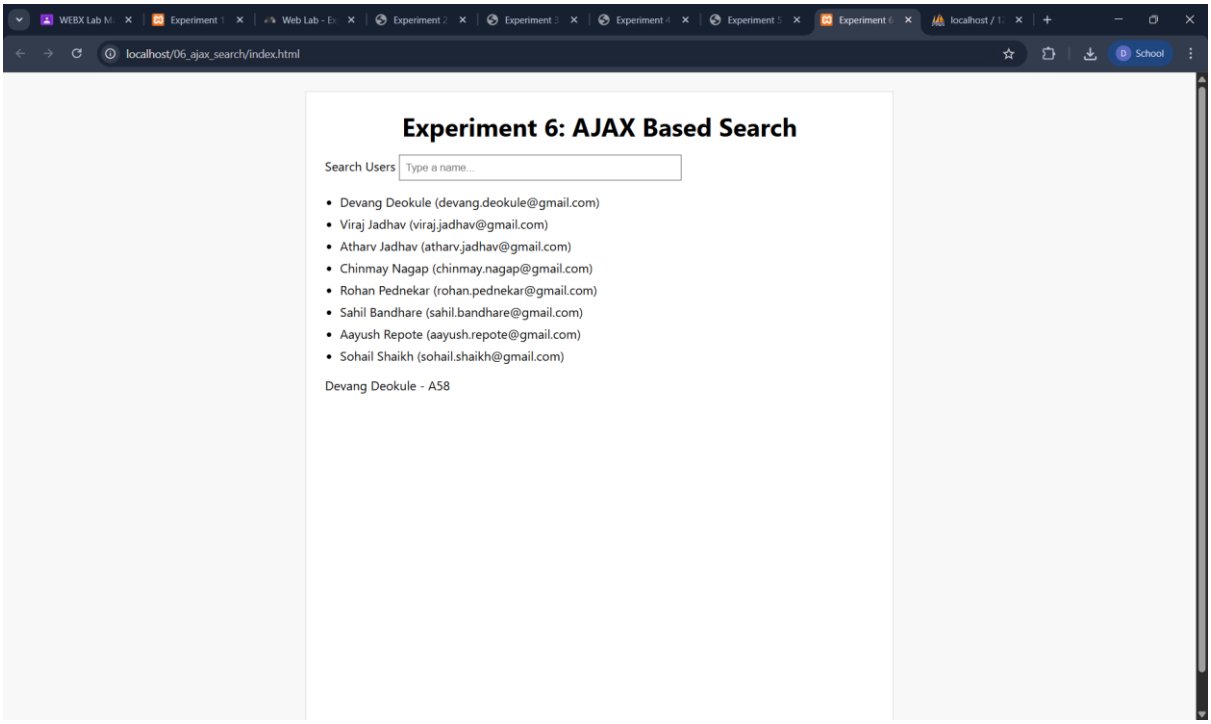
users.xml:

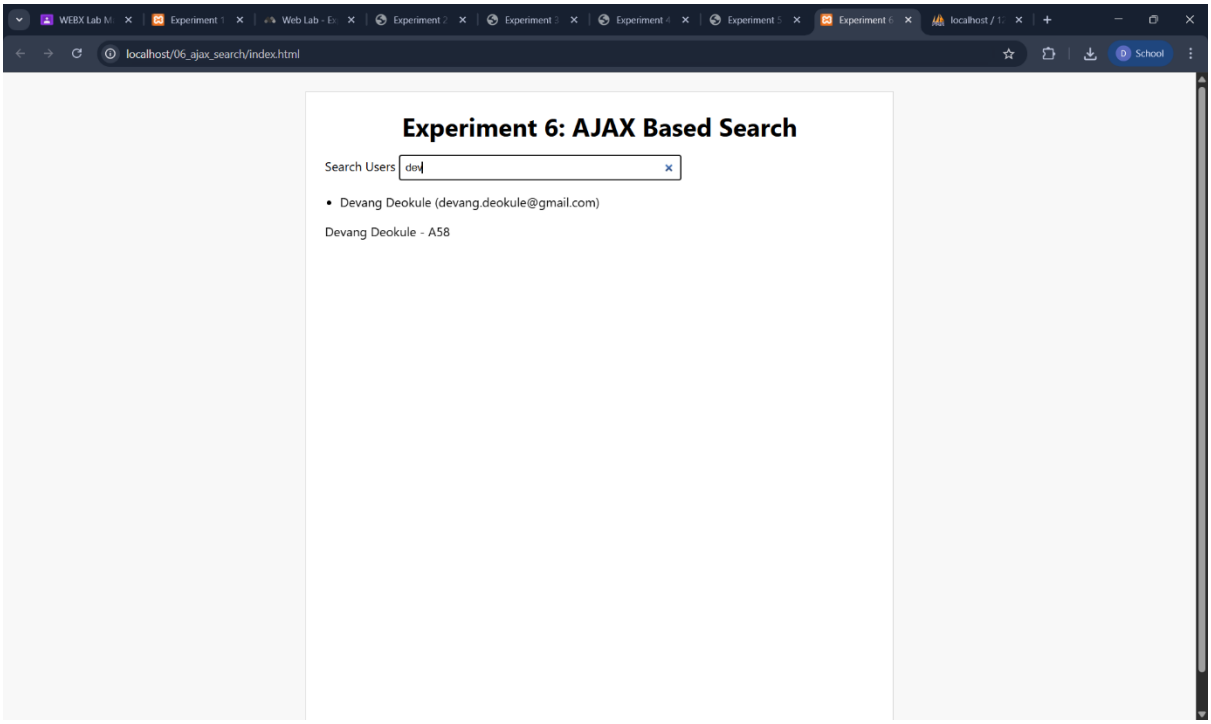
```
<?xml version="1.0" encoding="UTF-8"?>
<users>
  <user>
    <name>Devang Deokule</name>
    <email>devang.deokule@gmail.com</email>
  </user>
  <user>
    <name>Viraj Jadhav</name>
    <email>viraj.jadhav@gmail.com</email>
  </user>
  <user>
    <name>Atharv Jadhav</name>
    <email>atharv.jadhav@gmail.com</email>
  </user>
  <user>
    <name>Chinmay Nagap</name>
    <email>chinmay.nagap@gmail.com</email>
  </user>
  <user>
    <name>Rohan Pednekar</name>
    <email>rohan.pednekar@gmail.com</email>
  </user>
  <user>
    <name>Sahil Bandhare</name>
    <email>sahil.bandhare@gmail.com</email>
  </user>
  <user>
    <name>Aayush Repote</name>
    <email>aayush.repote@gmail.com</email>
  </user>
  <user>
    <name>Sohail Shaikh</name>
    <email>sohail.shaikh@gmail.com</email>
  </user>
</users>
```

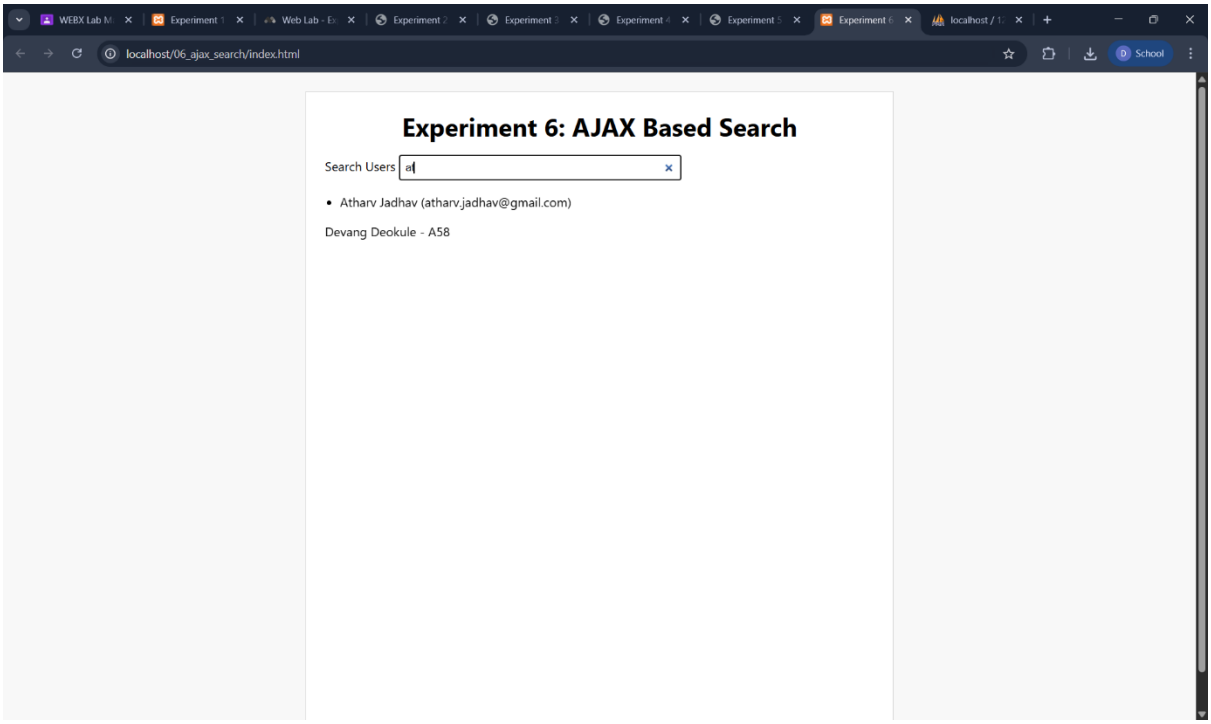
styles.css:

```
body {
  margin: 0;
  font-family: "Segoe UI", sans-serif;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  padding: 1.5rem 1rem;
  background: #f8f8f8;
}
.container {
  width: min(100%, 700px);
  padding: 1.5rem;
  border: 1px solid #d8d8d8;
  background: #fff;
}
h1 {
  max-width: 30ch;
  margin: 0 auto 1rem;
  text-align: center;
  font-weight: 700;
  line-height: 1.25;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
}
input {
  width: 100%;
  max-width: 360px;
  padding: 0.48rem;
  border: 1px solid #555;
}
ul {
  margin-top: 1rem;
  padding-left: 1.2rem;
}
li {
  margin: 0.4rem 0;
}
```

Output:




localhost/06_ajax_search/index.html
Experiment 6: AJAX Based Search
Search Users
dev
Devang Deokule (devang.deokule@gmail.com)
Devang Deokule - A58


localhost/06_ajax_search/index.html
Experiment 6: AJAX Based Search
Search Users
at
Atharv Jadhav (atharv.jadhav@gmail.com)
Devang Deokule - A58